

ACO TSP Solver

Ant Colony Optimization for the Travelling Salesman Problem

Artificial ants find short round trips through real Bosnian towns, live on screen

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Numerical Optimization

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Available on:

github.com/arminn2206/ProjNO_TSPACO_Memisevic

How It Works

A colony of virtual ants learns a short round trip, and you watch it happen.

- 1 The app picks **15–25 real Bosnian towns**. Goal: the shortest trip that visits each town once and returns home.
- 2 Each round, **virtual ants** build trips, preferring short hops and links other ants liked.
- 3 Shorter trips leave **more pheromone**; old pheromone **evaporates**. The colony converges on good routes.
- 4 Every result is measured against a **greedy nearest-neighbour route**.

TEST RESULTS · 30 INSTANCES · DEFAULT SETTINGS

0.68%

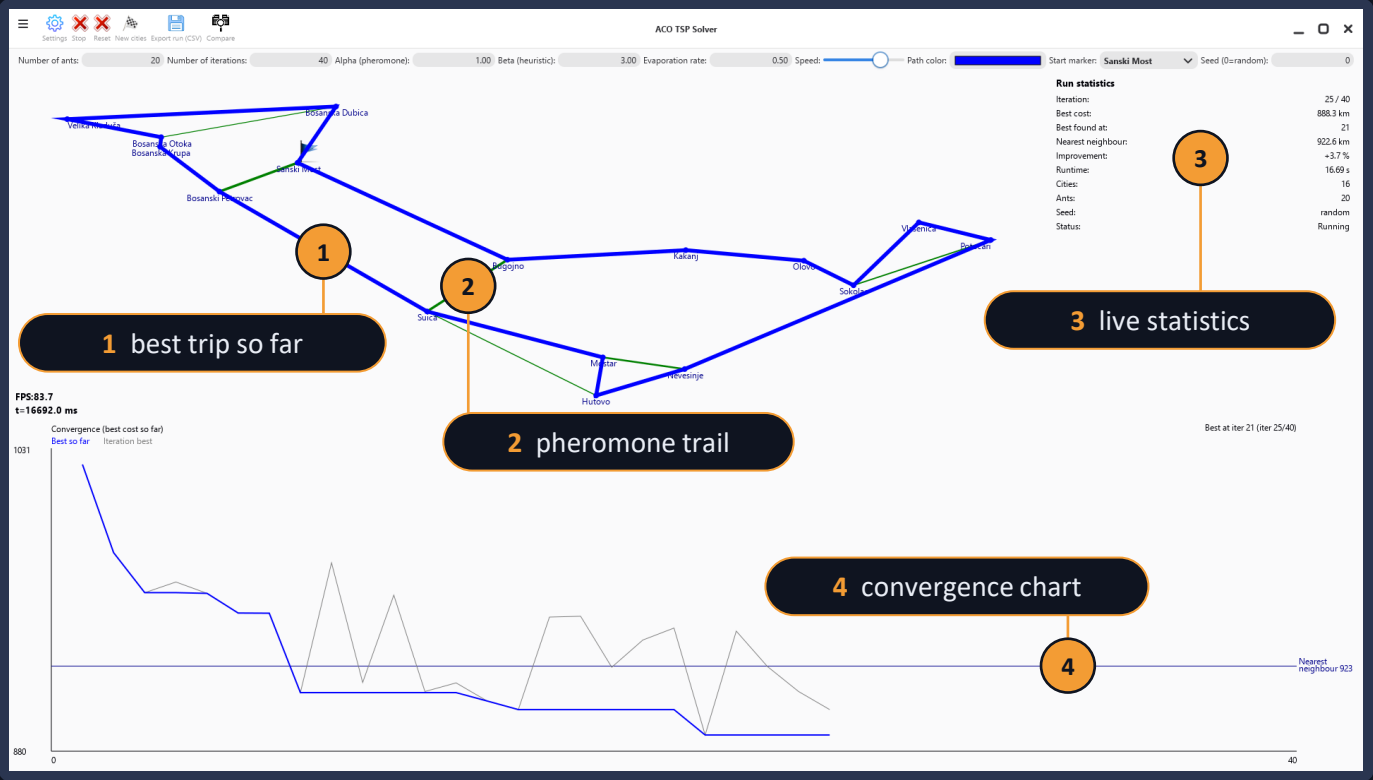
avg. above optimum

10 / 18

optimum found

0 / 30

worse than greedy



Live run: 16 towns, iteration 25 of 40. Best trip 888.3 km, 3.7% shorter than the greedy route (922.6 km).

Test figures: reference re-implementation, 30 instances; optimum computed exactly for the 18 with ≤ 20 towns.



Features & Controls

Everything from the proposal, plus tools to experiment and compare.

- ✓ **Real maps:** 15–25 towns drawn from 99 real Bosnian towns
- ✓ **Ant System:** basic Ant Colony Optimization, no shortcuts or local search
- ✓ **Parameters:** ants, iterations, α , β , evaporation, seed
- ✓ **Live map:** best trip + pheromone trail as line thickness
- ✓ **Convergence chart:** best so far, per-iteration best, greedy reference
- ✓ **Statistics:** iteration, best cost, improvement, runtime
- ✓ **Reproducible runs:** fixed seed replays a run on the same platform
- ✓ **Languages:** English and Bosnian
- ✓ **Platforms:** Windows, macOS and Linux

- ✓ **Start / Stop / Reset** run control
- ✓ **New cities** fresh instance anytime
- ✓ **Speed slider** watch slow or instantly
- ✓ **Compare** last two completed runs
- ✓ **Export (CSV)** run data + full instance
- ✓ **Settings** language, sound, trail display



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